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A systematic review protocol to explore goal planning in mental health service delivery

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Title: A systematic review protocol to explore goal planning in mental health service delivery.

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ABSTRACT

Introduction

Goal planning is widely recognised as an integral part of mental health service delivery and an important element in supporting recovery. Goal planning identifies priorities for treatment through discussion and negotiation between service users and health practitioners. Goal planning enhances motivation, directs effort, and focuses the development of strategies and treatment options to improve recovery outcomes and promote service users' ownership of the recovery process. Whilst goal planning is a common practice in mental health settings, evidence regarding its impact on treatment outcomes is lacking. This paper outlines a protocol for a systematic review that aims to explore the types of goals planned, experiences of service users and practitioners and the effectiveness of goal planning as a mental health intervention.

Methods and analysis

A systematic search will be conducted by searching Medline, CINAHL, Embase, Scopus, and PsycINFO electronic databases to answer the following questions: i) What types of goals are being developed within mental health care?; ii) What is the evidence for the effectiveness of goal planning on health and wellbeing for mental health service users?; iii) What are the experiences of mental health service users and their treating healthcare practitioners in relation to goal planning?; and iv) What are the barriers and facilitators to effective goal planning in mental health settings? Two independent researchers will screen the articles, selecting literature that meets criteria. All literature, regardless of study design that involves adult participants, with a mental illness and reporting on goal planning will be considered for inclusion. Data will be extracted from all eligible articles regardless of study design and summarised in a table. Appropriate quality assessment and data synthesis methods will be determined based on included study designs.

Ethics and dissemination

No ethics approval is required. The results will be disseminated through peer-reviewed publications and conference presentations.

PROSPERO registration number

Protocol has been submitted (ID 220595) and awaiting registration.

Strength and limitations of this study

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- This is the first systematic review focusing on the types of goals being set, experiences of service users and practitioners, and the effectiveness of goal planning as a mental health intervention.
- Study selection will be undertaken by two researchers to ensure all relevant studies are included without the risk of personal biases.
- All relevant studies will be included in the review, regardless of study design, to provide a deeper understanding of the impact of contextual factors and to map diverse measurements of effect.
- Appropriate synthesis of the data may be challenging given the diversity in study designs but will allow for a holistic understanding of the topic through the integration of findings from multiple sources of data.
- This review will only include studies published in English which may result in cultural and language bias.

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INTRODUCTION

Identifying and writing a goal is a familiar process to most people, improving the likelihood of achieving a desired outcome, and assisting us to accomplish and achieve throughout our lives.(1, 2) Goal planning has an important role in the rehabilitation of adults with a disability (e.g. following brain injury, mental illness or muscular injuries).(3) Goal planning within healthcare and rehabilitation settings has been found to have a positive effect on health-related quality of life, self-reported emotional status and service user self-efficacy.(3) A number of psychological theories underpin the process of goal planning and describe how setting goals allows people to monitor, alter or adapt their behaviour.(4) Social cognitive theory (5) emphasises the influence of self-efficacy on an individual's motivation to set and achieve goals, while health action theory (6) highlights the phases involved in goal planning and achievement. Goal setting theory (7) identifies the factors which impact on the effectiveness of goal planning including the specificity and difficulty of goals set and the influence of goal commitment, self-efficacy, task complexity and performance feedback on goal effects.

Goal planning is generally understood as a process of discussion and negotiation through which service users and health practitioners identify priorities for treatment to achieve a desired future state.(8) How goals are used in healthcare and rehabilitation settings is not always clearly defined and a range of terminology and approaches to goal planning have been described within the literature.(3) For the purpose of this review, the term goal planning will be used as it is commonly utilised in mental health recovery literature, encompassing other terms including goals, goal setting and goal attainment. To support positive outcomes, it is generally agreed that goals should be collaboratively developed, relevant to the person concerned, challenging but realistic and achievable, and include a component of measurability.(9, 10) Goal planning is used extensively within many healthcare settings and the use of goals is central in mental health service delivery and is a well-regarded component of psychiatric rehabilitation.(11)

Recovery is now an established concept that underpins international mental health policy and service provision and was developed by service users to counter the dominant biomedical discourse of chronic, enduring mental illness.(12, 13) While there is no universally accepted definition of recovery, it has been described as "a deeply personal, unique process of changing one's attitudes, values, feelings, goals, skills, and/or roles … a way of living a satisfying, hopeful, and contributing life even with limitations caused by illness."(14) This definition of recovery contrasts strongly with more traditional medical models of recovery that focus on outcomes such as decreased symptomatology, hospitalisations and medications.(15) Due to the influence of recovery, goal planning for people

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experiencing mental illness is likely to be different from other healthcare settings. Recovery-oriented practitioners aim to support a person to achieve their personal aims, desires, hopes and dreams in life, assisting them to identify goals or outcomes and preferred interventions to achieve these.(16) The pursuit of personal goals is an important recovery strategy and it is therefore essential that goal planning be considered in relation to psychiatric rehabilitation.(17)

While considerable attention has been given to researching the process of goal planning in rehabilitation generally,(3, 10, 18) there has been limited review of the literature regarding the use of goal planning in mental health service delivery. Jørgensen and Rendtorff (19) identified shared decision making and equal collaboration between mental health practitioners and service users as important in the setting of recovery-oriented goals. In contrast to traditional medical or authoritative models of healthcare, shared decision making is a collaborative, dynamic and interactive process between equally involved parties, exchanging information to agree on treatment priorities and processes.(20) Service users who experience goal planning that is a mutual or shared process perceive their goals to be more relevant, have a better understanding of, and engagement in the rehabilitation process.(21, 22) Shared decision making processes have been found to result in higher self-efficacy in mental health service users and a more positive attitude towards medication.(20)

Whilst policies increasingly support the inclusion of service users in goal planning, implementation lags behind and concerns regarding how to set goals that service users find personally meaningful have been noted.(23, 24) Research suggests that although health practitioners generally support the principles of shared decision making, in practice they tend to focus on the reduction of symptoms, providing expert advice regarding diagnosis, course of illness and treatment options.(25, 26) In contrast, service users emphasise broad, long-term goals that express their hopes and aspirations, prioritising goals related to non-medical concerns such as managing relationships, emotions and practical challenges such as housing and employment.(23, 26, 27) In addition, if the service user is assessed as experiencing severe mental health symptoms they may be seen as lacking capacity, with practitioners less likely to involve them in shared decision making processes.(19, 28)

Whilst goal planning is increasingly being used within mental health service delivery, there is no clear understanding of the types of goals being set, experiences of service users and practitioners, or the effectiveness of goal planning as a mental health intervention. A systematic review can facilitate the identification of literature exploring the effect of goal planning in mental health service delivery, and the experiences of key stakeholders, thereby providing guidance on how goal planning should be

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used in mental health service delivery, identifying knowledge gaps, and providing recommendations for future research.

Research questions

The aim of this systematic review is to explore the literature pertaining to goal planning within mental health service delivery from the perspective of health practitioners and service users, specifically:

- What types of goals are being developed within mental health service delivery?
- Is goal planning effective in improving the health and wellbeing of mental health service users?
- What are the experiences of mental health service users and their treating healthcare practitioners in relation to goal planning?
- What are the barriers and facilitators to effective goal planning in mental health service delivery from the perspectives of health practitioners and service users?

METHODS AND ANALYSIS

The systematic review process has been developed in accordance with recommendations from the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P)(29), presented in the online supplementary file. The protocol has been registered with the International Prospective Register of Systematic Reviews (PROSPERO Number: XXXX).

Search strategy

Medline (Ebsco), CINAHL, Embase (Embase.com), Scopus, and PsycINFO (Ovid) electronic databases will be searched. Once eligible documents identified, Google Scholar will be used for forward/backward citation checking. An academic librarian will be consulted for the validation of the databases and the final search terms. Terms will be searched as free text terms or as index terms (e.g. Medical Subject Headings) where appropriate.

The search will target studies reporting on the use of goal planning with adult mental health service users using keywords and subject headings. Search terms are outlined below utilising the PICO framework (Table 1). The PICO framework is commonly used in evidence-based healthcare and was adapted and used to structure the search strategy (30). 'P' refers to the population (adults experiencing mental illness); 'I' refers to the intervention (goal planning); 'C' refers to comparison group (which is not relevant for this study); and 'O' refers to outcome. As this study is interested in understanding a range of outcomes, no search term limitations will be used.

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Table 1. Systematic review search terms

PICO	Search Terms
P	"mental illness*" OR "mental health*" OR "mental disorder*" OR "mental health condition*" OR "mental health diagnosis" OR "psychiatric illness*" OR "psychiatric disorder*" OR "psychiatric condition*" OR "psychiatric diagnosis" OR "mental health consumer*" OR "mental health service user*" NOT child* OR youth* OR adolescent*
I	AND "Goals" [MeSH] OR (Goal* AND (set* OR plan* OR attain* OR achieve* OR assess* OR orient* or commit*) OR "action plan" OR "recovery plan"
O	No search limitations imposed.

There will be no date restriction set for study publication or study timeframe and all mental health settings (clinical, inpatient, community, recovery-focused) will be considered. Searches will be limited to peer-reviewed journals with grey literature, commentaries, conference abstracts, editorial or opinion pieces excluded.

Eligibility criteria

To be included in this review, studies must :

(i) involve adult participants (≥ 18 years); AND

(ii) involve participants diagnosed with any mental illness or mental health condition (e.g. depression, schizophrenia); AND

(iii) involve any form of goal planning; AND

(iv) report on the impact of goal planning on health outcomes; AND/OR

(v) report on the experiences of health practitioners and/or services users involved in goal planning processes.

All studies meeting the above criteria and published in the English language, will be considered for inclusion, regardless of study design.

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Studies will be excluded if they focus on carer experiences rather than the service user or practitioner or if they include a mixed population (e.g. report on a range of disabilities) where data specific to those experiencing mental illness cannot be extracted

Screening

Database search results will be downloaded and saved into Endnote where duplicates will be removed, and titles screened for relevance by one member of the research team. Remaining articles will be shared with another member of the research team utilising the Covidence systematic review software program. Abstracts will be independently screened regarding eligibility for study inclusion by two members of the research team. Articles that are deemed irrelevant by both reviewers will be excluded, those with a discrepancy will be included in the full text search. Full texts will be obtained for all remaining articles and screened independently by two members of the research team to determine eligibility for study inclusion. Reasons for exclusion will be recorded and any disagreements will be discussed and resolved through discussion with a third member of the research team.

Data extraction

The following information from each eligible article will be extracted and recorded in a tabular format: author, publication year, country, study design, aims, study setting, sample characteristics (e.g. diagnosis, age, gender), goal planning method, healthcare practitioner population(s), description of intervention(s), goal planning outcomes measure(s), service user results, and practitioner results.

Quality assessment

As this systematic review aims to include all studies relevant to the research questions, it is anticipated that included literature will employ a broad range of study designs (e.g. qualitative, quantitative, and mixed method studies). The inclusion of a range of studies can provide a deeper understanding of the impact of contextual factors and allow for analysis of the diversity of effect across a range of populations, interventions and settings.(31) However, appraising the methodological quality of studies with diverse designs remains challenging.(32) As such, it is only after the completion of data extraction that identification of the most appropriate quality assessment tool(s) will be possible. The Mixed Methods Appraisal Tool (MMAT) may be utilised if the resulting literature includes diverse study designs, allowing for the use of one tool to critically appraise quantitative descriptive studies, RCTs, nonrandomised studies, qualitative studies and mixed methods studies.(31) Alternatively, if the included studies are largely of a quantitative or

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qualitative design, or include study designs not assessed by the MMAT, appropriate tools will be chosen to accurately assess these study designs (e.g. Critical Appraisal Skills Programme (CASP) tools).

Data synthesis

Initially, data will be extracted and presented in a tabular format for descriptive synthesis. As it is likely that studies identified for this review will include diverse methodologies, it is anticipated that a mixed methods synthesis will be required.(33) Mixed methods synthesis allows for a holistic understanding of the topic by integrating findings from multiple sources of data and study types.(34) As this review aims to explore several research questions, a parallel-results convergent design for synthesising the data is likely to be the most relevant.(31) This synthesis design allows qualitative and quantitative evidence to be analysed separately and then brought together at the discussion and recommendation stage.(31) Quantitative and qualitative data synthesis methods will be informed by the type of studies meeting the eligibility inclusion criteria. The research team will discuss which methods best suit the included studies and which answer the research questions prior to data synthesis, and it is anticipated that a range of methodologies may be needed.

PATIENT AND PUBLIC INVOLVEMENT

This systematic review was developed and co-authored by a research team including health services researchers, health practitioners and a lived experience researcher, thereby integrating a range of expertise and perspectives.

ETHICS AND DISSEMINATION

Ethics approval is not required for this study as only published studies with non-identifiable data will be used. The results of the systematic review will be disseminated through conference presentations and a peer-reviewed manuscript.

CONCLUSION

Although goal planning is used extensively within mental health service delivery, there is limited understanding of the types of goals being set, experiences of service users and practitioners, and the effectiveness of goal planning as a mental health intervention. This systematic review will explore, present and analyse the evidence relating to the use of goal planning in mental health service delivery, thereby providing guidance on how goal planning should be used in mental health service delivery and highlighting gaps in the current literature.

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Contributions

All authors have contributed to the manuscript. The protocol was developed by VS, SM and AJW with HR reviewing the research questions, search strategy and terminology. VS wrote the first draft of the manuscript. SM, AJW, HR, SE-D and RN contributed to writing and reviewing the manuscript. All authors have reviewed and approved the final version of the manuscript and will be involved in conducting the systematic review.

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Competing interests

None declared.

Patient consent for publication

Not required.

Ethics approval

This study does not require formal ethical approval as primary data will not be collected. Results will be disseminated through a peer-reviewed publication and relevant conference presentations.

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4-6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	2
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6-8
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	7
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7-8
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7-8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8-9
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	9

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8-9
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	

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BMJ Open

A systematic review protocol to explore goal planning in mental health service delivery

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Title: A systematic review protocol to explore goal planning in mental health service delivery.

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ABSTRACT

Introduction

Goal planning is widely recognised as an integral part of mental health service delivery and an important element in supporting recovery. Goal planning identifies priorities for treatment through discussion and negotiation between service users and health practitioners. Goal planning enhances motivation, directs effort, and focuses the development of strategies and treatment options to improve recovery outcomes and promote service users' ownership of the recovery process. Whilst goal planning is a common practice in mental health settings, evidence regarding its impact on treatment outcomes is lacking. This paper outlines a protocol for a systematic review that aims to explore the types of goals planned, experiences of service users and practitioners and the effectiveness of goal planning as a mental health intervention.

Methods and analysis

A systematic search will be conducted during March 2021 by searching Medline, CINAHL, Embase, Scopus, and PsycINFO electronic databases to answer the following questions: i) What types of goals are being developed within mental health care?; ii) What is the evidence for the effectiveness of goal planning on health and wellbeing for mental health service users?; iii) What are the experiences of mental health service users and their treating healthcare practitioners in relation to goal planning?; and iv) What are the barriers and facilitators to effective goal planning in mental health settings? Two independent researchers will screen the articles, selecting literature that meets criteria. All literature, regardless of study design that involves adult participants, with a mental illness and reporting on goal planning will be considered for inclusion. Data will be extracted from all eligible articles regardless of study design and summarised in a table. Appropriate quality assessment and data synthesis methods will be determined based on included study designs.

Ethics and dissemination

No ethics approval is required. The results will be disseminated through peer-reviewed publications and conference presentations.

PROSPERO registration number CRD42020220595

Strength and limitations of this study

- This is the first systematic review focusing on the types of goals being set, experiences of service users and practitioners, and the effectiveness of goal planning as a mental health intervention.
- Study selection will be undertaken by two researchers to ensure all relevant studies are included without the risk of personal biases.

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- All relevant studies will be included in the review, regardless of study design, to provide a deeper understanding of the impact of contextual factors and to map diverse measurements of effect.
- Appropriate synthesis of the data may be challenging given the diversity in study designs but will allow for a holistic understanding of the topic through the integration of findings from multiple sources of data.
- This review will only include studies published in English which may result in cultural and language bias.

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INTRODUCTION

Identifying and writing a goal is a familiar process to most people, improving the likelihood of achieving a desired outcome, and assisting us to accomplish and achieve throughout our lives.(1, 2) Goal planning has an important role in the rehabilitation of adults with a disability (e.g. following brain injury, mental illness or muscular injuries).(3) Goal planning within healthcare and rehabilitation settings has been found to have a positive effect on health-related quality of life, self-reported emotional status and service user self-efficacy.(3) A number of psychological theories underpin the process of goal planning and describe how setting goals allows people to monitor, alter or adapt their behaviour.(4) Social cognitive theory (5) emphasises the influence of self-efficacy on an individual's motivation to set and achieve goals, while health action theory (6) highlights the phases involved in goal planning and achievement. Goal setting theory (7) identifies the factors which impact on the effectiveness of goal planning including the specificity and difficulty of goals set and the influence of goal commitment, self-efficacy, task complexity and performance feedback on goal effects.

Goal planning is generally understood as a process of discussion and negotiation through which service users and health practitioners identify priorities for treatment to achieve a desired future state.(8) How goals are used in healthcare and rehabilitation settings is not always clearly defined and a range of terminology and approaches to goal planning have been described within the literature.(3) For the purpose of this review, the term goal planning will be used as it is commonly utilised in mental health recovery literature, encompassing other terms including goals, goal setting and goal attainment. To support positive outcomes, it is generally agreed that goals should be collaboratively developed, relevant to the person concerned, challenging but realistic and achievable, and include a component of measurability.(9, 10) Goal planning is used extensively within many healthcare settings and the use of goals is central in mental health service delivery and is a well-regarded component of psychiatric rehabilitation.(11)

Recovery is now an established concept that underpins international mental health policy and service provision and was developed by service users to counter the dominant biomedical discourse of chronic, enduring mental illness.(12, 13) While there is no universally accepted definition of recovery, it has been described as "a deeply personal, unique process of changing one's attitudes, values, feelings, goals, skills, and/or roles … a way of living a satisfying, hopeful, and contributing life even with limitations caused by illness."(14) This definition of recovery contrasts strongly with more traditional medical models of recovery that focus on outcomes such as decreased symptomatology, hospitalisations and medications.(15) Due to the influence of recovery, goal planning for people

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experiencing mental illness is likely to be different from other healthcare settings. Recovery-oriented practitioners aim to support a person to achieve their personal aims, desires, hopes and dreams in life, assisting them to identify goals or outcomes and preferred interventions to achieve these.(16) The pursuit of personal goals is an important recovery strategy and it is therefore essential that goal planning be considered in relation to psychiatric rehabilitation.(17)

While considerable attention has been given to researching the process of goal planning in rehabilitation generally,(3, 10, 18) there has been limited review of the literature regarding the use of goal planning in mental health service delivery. Jørgensen and Rendtorff (19) identified shared decision making and equal collaboration between mental health practitioners and service users as important in the setting of recovery-oriented goals. In contrast to traditional medical or authoritative models of healthcare, shared decision making is a collaborative, dynamic and interactive process between equally involved parties, exchanging information to agree on treatment priorities and processes.(20) Service users who experience goal planning that is a mutual or shared process perceive their goals to be more relevant, have a better understanding of, and engagement in the rehabilitation process.(21, 22) Shared decision making processes have been found to result in higher self-efficacy in mental health service users and a more positive attitude towards medication.(20)

Whilst policies increasingly support the inclusion of service users in goal planning, implementation lags behind and concerns regarding how to set goals that service users find personally meaningful have been noted.(23, 24) Research suggests that although health practitioners generally support the principles of shared decision making, in practice they tend to focus on the reduction of symptoms, providing expert advice regarding diagnosis, course of illness, and treatment options.(25, 26) In contrast, service users emphasise broad, long-term goals that express their hopes and aspirations, prioritising goals related to non-medical concerns such as managing relationships, emotions and practical challenges such as housing and employment.(23, 26, 27) In addition, if the service user is assessed as experiencing severe mental health symptoms they may be seen as lacking capacity, with practitioners less likely to involve them in shared decision making processes.(19, 28)

Whilst goal planning is increasingly being used within mental health service delivery, there is no clear understanding of the types of goals being set, experiences of service users and practitioners, or the effectiveness of goal planning as a mental health intervention. A systematic review can facilitate the identification of literature exploring the effect of goal planning in mental health service delivery, and the experiences of key stakeholders, thereby providing guidance on how goal planning should be

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used in mental health service delivery, identifying knowledge gaps, and providing recommendations for future research.

Research questions

The aim of this systematic review is to explore the literature pertaining to goal planning within mental health service delivery from the perspective of health practitioners and service users, specifically:

- What types of goals are being developed within mental health service delivery?
- Is goal planning effective in improving the health and wellbeing of mental health service users?
- What are the experiences of mental health service users and their treating healthcare practitioners in relation to goal planning?
- What are the barriers and facilitators to effective goal planning in mental health service delivery from the perspectives of health practitioners and service users?

For the purpose of the review, the definition of a rehabilitation goal by Siegert and Levack will be applied i.e. “Rehabilitation goals are actively selected, intentionally created, have purpose and are shared (where possible) by the people participating in the activities and interventions designed to address the consequence of acquired disability” with goal planning defined as “the establishment or negotiation of rehabilitation goals”.(8 p11)

METHODS AND ANALYSIS

The systematic review process has been developed in accordance with recommendations from the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P)(29), presented in the online supplementary file. The protocol has been registered with the International Prospective Register of Systematic Reviews (PROSPERO Number: CRD42020220595).

Search strategy

Medline (Ebsco), CINAHL, Embase (Embase.com), Scopus, and PsycINFO (Ovid) electronic databases will be searched in March 2021. Once eligible documents identified, Google Scholar will be used for forward/backward citation checking. An academic librarian will be consulted for the validation of the databases and the final search terms. Terms will be searched as free text terms or as index terms (e.g. Medical Subject Headings) where appropriate.

The search will target studies reporting on the use of goal planning with adult mental health service users using keywords and subject headings. Search terms are outlined below utilising the PICO framework (Table 1). The PICO framework is commonly used in evidence-based healthcare and was

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adapted and used to structure the search strategy (30). 'P' refers to the population (adults experiencing mental illness); 'I' refers to the intervention (goal planning); 'C' refers to comparison group (which is not relevant for this study); and 'O' refers to outcome. As this study is interested in understanding a range of outcomes, no search term limitations will be used.

Table 1. Systematic review search terms

PICO	Search Terms
P	"mental disorder"[MeSH] OR "mental health"[MeSH] OR "mental illness*" OR "mental health*" OR "mental disorder*" OR "mental health condition*" OR "mental health diagnosis" OR "psychiatric illness*" OR "psychiatric disorder*" OR "psychiatric condition*" OR "psychiatric diagnosis" OR "mental health consumer*" OR "mental health service user*" NOT child* OR youth* OR adolescent*
I	AND ("Goals" [MeSH] OR Goal*) OR ("Patient Care Planning"[MeSH] OR (Care AND plan*)) OR ("Decision making, shared"[MeSH] OR "shared decision mak*") AND (set* OR plan* OR attain* OR achieve* OR assess* OR orient* OR commit* OR "action plan" OR "recovery plan") Note: search terms will be modified to suit individual databases e.g., Emtree or Subject Headings
O	No search limitations imposed.

There will be no date restriction set for study/literature publication or study timeframe and all mental health settings (clinical, inpatient, community, recovery-focused) will be considered. Searches will be limited to peer-reviewed journals with grey literature, commentaries, conference abstracts, editorial or opinion pieces excluded.

Eligibility criteria

To be included in this review, studies must :

(i) involve adult participants (≥ 18 years); AND

(ii) involve participants diagnosed with any mental illness or mental health condition (e.g. depression, schizophrenia); AND

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(iii) involve any form of goal planning; AND

(iv) report on the impact of goal planning on health outcomes; AND/OR

(v) report on the experiences of health practitioners and/or services users involved in goal planning processes.

All studies meeting the above criteria and published in the English language, will be considered for inclusion, regardless of study design.

Studies will be excluded if they focus on carer experiences rather than the service user or practitioner or if they include a mixed population (e.g. report on a range of disabilities) where data specific to those experiencing mental illness cannot be extracted

Screening

Database search results will be downloaded and saved into Endnote where duplicates will be removed, and titles screened for relevance by one member of the research team. Remaining articles will be shared with another member of the research team utilising the Covidence systematic review software program. Abstracts will be independently screened regarding eligibility for study inclusion by two members of the research team. Articles that are deemed irrelevant by both reviewers will be excluded, those with a discrepancy will be included in the full text search. Full texts will be obtained for all remaining articles and screened independently by two members of the research team to determine eligibility for study inclusion. Reasons for exclusion will be recorded and any disagreements will be discussed and resolved through discussion with a third member of the research team.

Data extraction

The following information from each eligible article will be extracted and recorded in a tabular format: author, publication year, country, study design, aims, study setting, sample characteristics (e.g. diagnosis, age, gender), goal planning method, healthcare practitioner population(s), description of intervention(s), goal planning outcomes measure(s), service user results, and practitioner results.

Quality assessment

As this systematic review aims to include all studies relevant to the research questions, it is anticipated that included literature will employ a broad range of study designs (e.g. qualitative, quantitative, and mixed method studies). The inclusion of a range of studies can provide a deeper understanding of the impact of contextual factors and allow for analysis of the diversity of effect

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across a range of populations, interventions and settings.(31) However, appraising the methodological quality of studies with diverse designs remains challenging.(32) As such, it is only after the completion of data extraction that identification of the most appropriate quality assessment tool(s) will be possible. The Mixed Methods Appraisal Tool (MMAT) may be utilised if the resulting literature includes diverse study designs, allowing for the use of one tool to critically appraise quantitative descriptive studies, RCTs, nonrandomised studies, qualitative studies and mixed methods studies.(31) Alternatively, if the included studies are largely of a quantitative or qualitative design, or include study designs not assessed by the MMAT, appropriate tools will be chosen to accurately assess these study designs (e.g. Critical Appraisal Skills Programme (CASP) tools).

Data synthesis

Initially, data will be extracted and presented in a tabular format for descriptive synthesis. As it is likely that studies identified for this review will include diverse methodologies, it is anticipated that a mixed methods synthesis will be required.(33) Mixed methods synthesis allows for a holistic understanding of the topic by integrating findings from multiple sources of data and study types.(34) As this review aims to explore several research questions, a parallel-results convergent design for synthesising the data is likely to be the most relevant.(31) This synthesis design allows qualitative and quantitative evidence to be analysed separately and then brought together at the discussion and recommendation stage.(31) Quantitative and qualitative data synthesis methods will be informed by the type of studies meeting the eligibility inclusion criteria. The research team will discuss which methods best suit the included studies and which answer the research questions prior to data synthesis, and it is anticipated that a range of methodologies may be needed. The literature review and synthesis will be completed in 2021.

PATIENT AND PUBLIC INVOLVEMENT

This systematic review was developed and co-authored by a research team including health services researchers, health practitioners and a lived experience researcher, thereby integrating a range of expertise and perspectives.

ETHICS AND DISSEMINATION

Ethics approval is not required for this study as only published studies with non-identifiable data will be used. The results of the systematic review will be disseminated through conference presentations and a peer-reviewed manuscript.

CONCLUSION

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Although goal planning is used extensively within mental health service delivery, there is limited understanding of the types of goals being set, experiences of service users and practitioners, and the effectiveness of goal planning as a mental health intervention. This systematic review will explore, present and analyse the evidence relating to the use of goal planning in mental health service delivery, thereby providing guidance on how goal planning should be used in mental health service delivery and highlighting gaps in the current literature.

Acknowledgements

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Contributions

All authors have contributed to the manuscript. The protocol was developed by VS, SM and AJW with HR reviewing the research questions, search strategy and terminology. VS wrote the first draft of the manuscript. SM, AJW, HR, SE-D and RN contributed to writing and reviewing the manuscript. All authors have reviewed and approved the final version of the manuscript and will be involved in conducting the systematic review.

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Competing interests

None declared.

Patient consent for publication

Not required.

Ethics approval

This study does not require formal ethical approval as primary data will not be collected. Results will be disseminated through a peer-reviewed publication and relevant conference presentations.

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PRISMA-P 2015 Checklist

This checklist has been adapted for use with protocol submissions to *Systematic Reviews* from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	x		43-45
Update	1b	If the protocol is for an update of a previous systematic review, identify as such			n/a
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	x		61
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	x		3-29
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	x		263-267
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments			n/a
Support					
Sources	5a	Indicate sources of financial or other support for the review	x		269-270
Sponsor	5b	Provide name for the review funder and/or sponsor			n/a
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol			n/a
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	x		138-144
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	x		146-160
METHODS					

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	x		181-198
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	x		167-171
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	x		179-180
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	x		200-203
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	x		200-209
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	x		211-215
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	x		211-215
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	x		211-215
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	x		217-230
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	x		232-243
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	x		232-243
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	x		232-243
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	x		232-243
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)		x	
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)		x	

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